



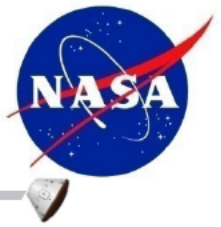
OCADS and FOD Disciplines

***JSC Flight Operations Directorate
Flight Integration Division
Flight Ops Safety Office***

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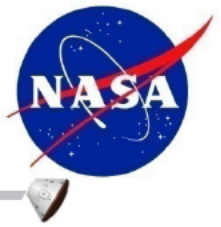
July 26, 2022

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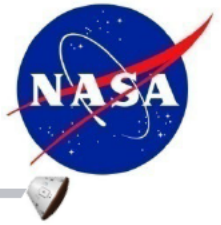
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Brief OCAD Refresher



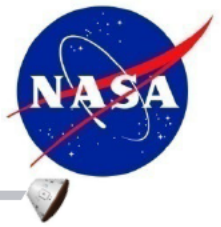
- An *operations (ops) control* is the least-desirable form of hazard control
- A Hazard Report author writes a proposed ops control to the OCAD database
- FOD Subject Matter Experts review and disposition the OCAD
 - Nominally this is a 30-day template
 - » Occasionally, the ops tempo makes this 30-day window difficult
 - OCADs are Approved, Approved with Mod, Rejected
 - » Once an OCAD is approved or approved with mod, that control in the hazard report can be marked verified
 - » OCADs that are rejected can be appealed or rewritten by author
- FOD Ops has a 'road show' on writing effective ops controls which can be presented upon request

FOD Concerns with OCADs



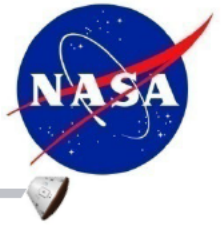
- FOD Ops Disciplines devote significant effort in the review of proposed OCADs
- A recent evaluation showed that a majority of proposed OCADs required significant back-and-forth iterations
- FOD disciplines have rejected very few OCADs
 - Instead, FOD disciplines work with authors to fill in details
 - Clarify locations, part numbers, ops nom, etc
 - With volume of OCADs and required clarifications, this is an unacceptable workload on FOD disciplines
- FOD Ops Safety concurs with the ops disciplines to reject OCADs that don't meet some minimum content standards

OCAD Expectations



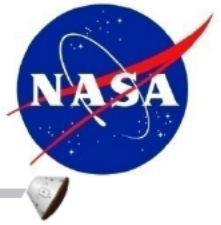
- Note – this is a guide that FOD Ops Safety will use to pre-screen OCADs prior to distributing them to the Ops Disciplines for evaluation
- 1. Clearly identify the hazard being controlled
 - Why do we need the hazard control?
 - Identify when the control applies, and to what hardware
 - Include reference to HR and cause/control number
 - “To control XXX, do YYY” type of language
- 2. The OCAD must stand alone
 - The Ops Disciplines are not going to dig through hazard reports to understand the context of the OCAD
 - The “Additional Information” field is a great place to add context
 - Between the OCAD language and the Additional Information field, the OCAD must make sense
 - Attached tables and drawings (e.g. proposed no-touch areas, cable mating table) are very valuable - attach them to the OCAD
 - » Ops Control wording must reference the attachment, e.g. “Refer to attached diagram of

OCAD Expectations (cont'd)



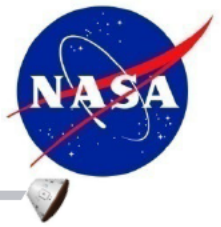
3. Ensure all pertinent information is present and correct
 - Op nom
 - Part number (s)
 - Location (s)
 - Torque values
 - » As a range, no reference to running torque
 - RPCM ID/location
 - Etc.
4. Ensure all inhibits are identified (by part number, location, etc)
5. Avoid writing the controls as procedure steps
 - FOD is responsible for writing the procedures to implement the OCAD
 - FOD discipline most likely has additional contextual information which is germane to the resulting procedures

Other OCAD 'Goodness'



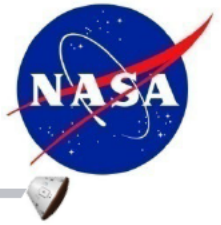
- Spell out acronyms on first use
- Ensure it's not a design control
 - “the lens is non-shatterable material” is design control
 - “When not in use, stow the lens to prevent breakage” IS ops control
- Use ‘impersonal’ statements
 - “... crew shall torque fastener ...” is bad
 - “... fasteners shall be torqued to ...” is good
- Don't state the obvious (e.g. “don't slam the door on your fingers”)

Summary



- **FOD's goal is to get Ops Controls evaluated / accepted in a timely fashion**
 - The 'template' goal is 30 working days
 - Priority is given to those OCADs that are needed soonest
 - Completion of an OCAD is not the end – ops procedures need to be written, tested, and approved
- **Workload on FOD Disciplines to evaluate OCADs can be significant**
 - Iterating back and forth to fill in details is not practicable
 - This is contributing to delays in OCAD evaluation
 - FOSO will pre-screen OCADs prior to the Ops Disciplines evaluation
 - » Incomplete or inadequate OCADs will not be distributed to the Ops Disciplines for evaluation
 - » FOSO and/or the Ops Disciplines will reject incomplete or inadequate OCADs

Summary (cont'd)



- **Notes:**
 - **FOSO can help pre-screen OCADs**
 - **Last-minute OCADs are to be avoided**
 - » **Any Ops Control that is required within 30 days should be done with a Chit**
 - » **Ops Disciplines do not have time to review and approve OCADs and then write and approve procedures in such a short time-frame**
 - **FOSO has 'road show' to help OCAD authors**